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Roll No.

220965B

6th Sem/ Electrical
Subject : Smart Grid & Distributed Generation System

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Smart Grid Technology comprises the following :

- a) AMI, PLM, OMS Renewable Integration
- b) DR/DSM, distribution Automation
- c) Energy Efficient Systems, Micro grid
- d) All of the above

Q.2 AMI means

- a) Automated Metering Instrument
- b) Alternate Metering Instrument
- c) Advanced Metering Infrastructure
- d) Advanced Metering Instrument

Q.3 What is the primary advantages of a smart Grid over a traditional grid?

- a) Higher transmission losses
- b) Improved efficiency and reliability
- c) Increased manual monitoring
- d) Higher carbon emissions

- Q.4 Which technology is commonly used in Smart Grids for two-way communication?
- a) SCADA
 - b) GSM
 - c) IoT
 - d) AMI
- Q.5 Which of the following is NOT a component of a Smart Grid?
- a) Smart meters
 - b) Fossil fuel power plants
 - c) Distributed energy resources
 - d) Energy storage systems
- Q.6 Demand Response in a Smart Grid helps to :
- a) Increase energy consumption
 - b) Balance supply and demand
 - c) Increase dependency on fossil fuels
 - d) Reduce grid automation

SECTION-B

- Q.7 SCADA is used for real-time monitoring and control in Smart Grids. (T/F)
- Q.8 Power Line Communication (PLC) is commonly used for Smart Grid communication. (T/F)
- Q.9 A smart Grid helps in reducing carbon emissions by integrating renewable energy. (T/F)

- Q.10 The main function of an inverter in a Distributed Generation system is to convert _____ current to _____ current.
- Q.11 One major challenge of integrating Distributed Generation into the power grid is _____.
- Q.12 Write the full-form of SCADA.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Discuss the evolution of electric grid system. (CO1)
- Q.14 What is the need of Smart grid. (CO1)
- Q.15 Discuss about plug in Hybrid Electric Vehicle (PHEVs) (CO2)
- Q.16 Draw the Architecture Smart grid. (CO2)
- Q.17 Explain the Architecture of SCADA. (CO2)
- Q.18 Discuss the smart energy System. (CO3)
- Q.19 What are the benefits of AMI? (CO3)
- Q.20 Write the overview of microgrid. (CO3)
- Q.21 What is the function of SCADA in Smart grid. (CO4)
- Q.22 Explain the standards for interconnecting distributed resources to, electric power systems. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain smart communication and smart management with respect to smart grid technology. (CO1)

Q.24 Explain Home Energy Management System. (CO2)

Q.25 Discuss the plug-in Hybrid electrical vehicle in detail. (CO4)